

March 14, 2025

Attn: Faisal D'Souza, NCO
AI Action Plan
2415 Eisenhower Avenue
Alexandria, VA 22314
Via email: [REDACTED]

Re: Schneider Electric Response to the Trump Administration's Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

To Whom It May Concern:

As a leading infrastructure provider for data centers, the grid, and other critical elements of the United States' energy infrastructure, Schneider Electric appreciates the opportunity to provide recommendations for the Trump Administration's AI Action Plan. To propel American leadership in AI, lower costs for consumers, boost economic growth and good jobs, and make the best use of American energy, it is critical the Administration take bold action. We are enthusiastic about the Administration's speed and attention to this issue and look forward to remaining a partner to the Administration as it crafts the AI Action Plan and implements its energy policy agenda.

Schneider Electric has been a leader in innovation in energy in the United States for nearly two centuries as a manufacturer, software company, and innovator employing over 21,000 Americans across more than 20 factories, facilities, and distribution centers across the country. Within the data center market, Schneider Electric serves as a leading physical infrastructure provider, providing the power and cooling systems that keep data center operations reliable and secure. In recent years, Schneider Electric has invested more than \$440 million to expand our U.S. capacity, adding to our American manufacturing footprint and bolstering the U.S. supply chain for critical grid and energy products and components – including those necessary to support the AI and data center boom. In fact, many of our recent investments in the U.S. are strengthening domestic supply of data center equipment to support the AI boom, including low- and medium-voltage switchgear and uninterruptible power supplies (UPSs). Driven by the market, an all-of-the-above energy strategy, and an increasingly flexible policy landscape, we are eager to continue growing in the U.S. to not only support the American energy supply chain, but to contribute to America's growing manufacturing base and highly skilled workforce.

Schneider Electric is unique in that we play a key role in every stage of the AI value chain: we provide critical power to semiconductor factories; we serve as manufacturers of critical data center equipment; we are consumers of AI technologies and solutions to optimize our industrial processes and business; and we are developers of AI software and solutions for our customers. As infrastructure providers for data centers, we provide uninterruptible power systems (UPSs), cooling equipment, racks, switchgear, and more that keep data centers running efficiently and

securely. Our software and solutions also play a role in keeping data centers operational, resilient, and secure. Together, our hardware and AI-driven software can help data centers meet the growing challenge of mounting and variable power consumption needs. Along with our partners, we are leading in deploying specific, AI-ready technologies that will power, cool, and support the most advanced and intensive workloads.¹

Our expansive software portfolio, including industrial software powered by AVEVA, a subsidiary of Schneider Electric, plays an integral role in the future of AI data centers and industrial AI that will drive the most competitive industry of the future. AVEVA operates its software across various sectors – including critical infrastructure and data centers – to improve efficiency, reduce costs, meet demand, optimize investments, and improve quality of life. Critically, AVEVA software provides key insights into entire value chains. In a data center context, this software can cultivate impactful data on the use of electricity in real-time and can share how each stakeholder – like the data center itself or a water utility – is impacting the demand and load curves. In the context of manufacturing, AVEVA can optimize the energy management and industrial processes, safety, and maintenance requirements of factories that make critical components required for AI supply chains and necessary for AI leadership.

Data cultivated via industrial software solutions can enable the modeling of digital twins, which is a virtual representation of a physical object, process, or system – including data center operations – to simulate the behavior and performance of its real-world counterpart in real-time. Leveraging digital twins for data centers allows for improved decision-making, optimized performance, and reduced downtime and maintenance costs. Simply put, using industrial software generates savings for power providers and data center operators. For example, Duke Energy saved over \$34 million in a single early-catch event in 2016 leveraging AVEVA software.² Additionally, Southern Company has leveraged AVEVA’s digital twin, AI-powered technology to unlock \$10 million in savings to date.³

The AI buildout will evolve rapidly, requiring an energy backbone capable of propelling it and a policy landscape flexible enough to shepherd innovation. In the coming years, Schneider Electric expects a new wave of AI buildout, wherein AI compute demands will shift from largely training models to working – or inference – models. Working AI applies its learned knowledge to make predictions or take actions on new data, requiring progressively more and more compute capacity as AI matures and becomes increasingly useful across the American economy. This pending shift will require speed and agility on the part of all stakeholders involved in the growth of the data center market in the U.S. From utilities and power providers to infrastructure manufacturers and policymakers, the need for collaborative action to continue driving and powering the AI boom is critical to sustain American AI leadership. Schneider Electric’s unique leadership, technical

¹ See Schneider Electric, *Schneider Electric Collaborates with NVIDIA on Designs for AI Data Centers* (Mar. 18, 2024) at <https://www.se.com/ww/en/about-us/newsroom/news/press-releases/schneider-electric-collaborates-with-nvidia-on-designs-for-ai-data-centers-65f801c8595e3ff8cd0a80c8> (announcing Schneider Electric’s collaboration with NVIDIA to optimize data center infrastructure by releasing AI data center reference designs to redefine the benchmarks for AI deployment and operations within data center ecosystems).

² See AVEVA, *Duke Energy Predictive Analytic Success Story* at https://www.aveva.com/en/perspectives/success-stories/duke-energy/?_gl=1*193pw09*_up*MQ..*_gs*MQ..&gclid=Cj0KCQjwhMq-BhCFARIsAGvo0Kfqdeu-nC9XFnc_DCsF1FNEHOWWObIG56cePWnwnNH6KjAZymKVmwcaAvtjEALw_wcB.

³ See AVEVA, *Southern Company* at <https://www.aveva.com/en/perspectives/success-stories/southern-company-digital-twin/>.

expertise, and solutions in energy management will play a vital role in building the backbone of AI infrastructure in the U.S.

As the Administration considers its AI policies, we stand ready to serve as a partner. We encourage attention to policies that bolster and modernize U.S. energy infrastructure, promote greater affordability and cost-containment, drive data center and grid flexibility, (including the deployment of distributed resources as stabilizers for critical infrastructure), and build resilient supply chains. We support the concerted deployment of readily available demand-side solutions at scale and more rapidly while concurrently developing a comprehensive national plan for supply-side solutions recognizing those solutions can vary in timeline to permitting, construction and power. Specifically, we encourage the Administration to enact a comprehensive strategy to power and unlock the future of AI in America including policy actions that:

Strengthen and Modernize the Energy Infrastructure Needed to Power AI

- Create a Grid Czar in the Trump Administration to coordinate a comprehensive national plan to modernize and bolster the grid for the AI Era, working effectively with all players in the ecosystem to break down unnecessary barriers and pave the way for success;
- Modernize, expand, and digitalize U.S. grid infrastructure, including by ensuring the effective dispersal and deployment of remaining grid modernization funds allocated under the Infrastructure Investment and Jobs Act (IIJA) to support a range of utility infrastructure upgrades alongside deployment of grid-edge technologies and software solutions;
- Ease and speed permitting processes for data centers and supporting infrastructure, including permits for high-voltage interstate transmission infrastructure;
- Streamline time to power by improving interconnection queues and establishing clear processes and criteria for prioritization that are transparent, timely, and support national, energy, and economic security objectives;
- Categorize data centers as critical infrastructure given the vital role they play in financial, economic, and national security, and;
- Develop clear regulatory guidance to facilitate the approval and development of co-located data centers with energy generation resources, including everything from interconnection processes to tariff design and cost allocation for necessary grid upgrades of transmission and distribution infrastructure.

Increase Data Center Flexibility and Improve Load Management to Keep Energy Costs Low and Reduce Grid Strain

- Provide guidance and direction to data center operators, power providers, and infrastructure providers regarding best practices for optimizing energy usage to maintain low costs and reliable energy access for electricity consumers;
- Support and incentivize data center flexibility with the grid to minimize impact to consumers, communities, and our energy infrastructure;
- Enact the policy, programs, and appropriations or incentives required to accelerate more rapid deployment at scale of stationary storage (batteries) and islandable-power via microgrids in critical infrastructure (data centers, military installations, hospitals, or

schools) to stabilize and harden the grid against cybersecurity and weather events that disrupt national and energy security, and;

- Promote government efficiency by leveraging the power of third-party private financing and public-private partnerships to scale investment in microgrids for data centers supporting the federal government, military, or public sector.

Strengthen the Supply of Energy Resources and Products Supporting the AI Boom

- Streamline permitting processes to speed the buildout of new mining, mining processing, manufacturing, and production of materials, components, and products needed to buildout the AI economy in the U.S.;
- Identify strategic supply chains critical to American AI leadership including advanced and legacy semiconductors, critical minerals, critical grid equipment, and batteries, and empower suppliers to produce these goods regionally;
- Preserve and enact effective incentives to drive investment in localized supply chain in key areas critical to AI, and;
- Adopt effective trade agreements or memorandums of understanding (MOUs) with international partners to ensure ample capacity and bridge supply gaps until localized supply chains are in place.

Drive Investment in the Future of AI in America

- Remove roadblocks for future investment in AI as an industry, ensuring capital can accelerate towards building AI as a differentiated capability of U.S. national security interest;
- Advance research and development (R&D) of new and emerging technologies that leverage AI or support AI infrastructure such as power electronics for UPS systems, smart building controls, long-duration energy storage (LDES), and grid-interactive capabilities and;
- Ensure AI regulation is innovation- and competition-friendly.

Advance Responsible AI Cybersecurity Policies

- Support the development of voluntary, consensus-based international standards, such as those developed in ISO/IEC;
- Encourage the adoption of techniques – including AI security risk scoring and management, threat modeling, adversarial tests, secure supply chain, and xBOM – that improve the security of AI, and;
- Advance policies that support AI-powered cybersecurity in critical infrastructure.

American AI policy that not only drives private sector innovation but seeks to power it will both surge American leadership in AI and ensure American energy independence and dominance.

Strengthen and Modernize the Energy Infrastructure Needed to Power AI

Unlocking America's AI future hinges on the strength and reliability of the U.S.'s power infrastructure. Today, our power infrastructure's reliability is being appropriately questioned in the face of an unprecedented amount of load growth and it is increasingly clear that our already-

constrained grid infrastructure requires expedient, strategic attention and investment to continue supporting American innovation and leadership. In our white paper, *The AI Disruption: Challenges and Guidance for Data Center Design*⁴, we estimate that AI currently represents 4.5 gigawatts (GW) of power consumption, and we project this to grow to a total consumption of 14 GW to 18.7 GW by 2028. Other projections estimate that data centers are expected to account for up to 44 percent of U.S. electricity load growth through 2028.⁵

Bolster the U.S. Grid

Importantly, overall increases in load growth are not solely associated with AI and data centers – other contributing factors include increased U.S. manufacturing and electrification demands. This reality underscores the need for a comprehensive strategy to address growing load demand, ensuring that America can lead on – and power – AI. While energy and AI stakeholders continue working together to innovate and collaboratively partner on solutions to address the energy crunch, policymakers must prioritize modern upgrades to U.S. power infrastructure to best position the U.S. for global AI leadership.

Continued and future investment in grid modernization will propel the entire data center market – irrespective of computing obligations – to be more efficient consumers of energy. Specifically, investments to strengthen both the transmission and distribution of electricity along the grid will ensure that the U.S.’s energy infrastructure is up to the task of supporting data center buildout.⁶ Additionally, grid-strengthening investments that digitalize grid infrastructure, add layers of resilience and flexibility through strategically placed distributed energy resources and microgrids, and speed permitting and interconnection processes to bring critical energy technologies and resources online faster will lay the foundation for American energy dominance, including our ability to power AI.

Policymakers should also look to AI-driven technologies to assist in optimizing and modernizing the grid. There are several use cases for AI in the energy sector and we are discovering more applications as we continue developing and innovating. For example, AI can be used to track, forecast, and manage increasing load demand, maintain assets to prevent costly equipment failures, leverage microgrids to allow critical facilities to “island” from the grid, and preserve and protect assets at water and wastewater facilities. Additionally, AI can be used in the industrial sector to generate energy cost savings by optimizing assets and providing early catch warnings.⁷

⁴ Victor Avelar, Patrick Donovan, Paul Lin, Wendy Torell, and Maria A. Torres Arango, *The AI Disruption: Challenges and Guidance for Data Center Design*, Schneider Electric – Energy Management Research Center, Sep. 7, 2023, at 2.

⁵ See Maeghan Rouch, Aaron Denman, Peter Hanbury, Paul Renno, and Ellyn Grey, *Utilities Must Reinvent Themselves to Harness the AI-Driven Data Center Boom*, Bain & Company (Oct. 2024) at <https://www.bain.com/insights/utilities-must-reinvent-themselves-to-harness-the-ai-driven-data-center-boom/> (citing data from Bank of America and Goldman Sachs analyst forecasts (April 2024); EIA Short-Term Energy Outlook (May 2024); EIA 2023 Outlook; IDC Datacenter Deployment and Spend Forecast (H2 2023); and Bain analysis).

⁶ To this point, we specifically encourage the Administration to preserve and advance funding through the Department of Energy’s Grid Deployment Office.

⁷ See AVEVA, *Duke Energy Predictive Analytic Success Story* at https://www.aveva.com/en/perspectives/success-stories/duke-energy/?_gl=1*193pw09*_up*MQ..*_gs*MQ..&gclid=Cj0KCQjwhMq-BhCFARIsAGvo0Kfqdeu-nC9XFnc_DCsF1FNEHOWWObIG56cePWnwnNH6KjAZymKVmwcaAvtjEALw_wcB.

Build Data Centers as Grid Stabilizers

Investments in the grid and supporting power infrastructure must run concurrently with investments to optimize and buildout AI data centers. As the data center market quickly evolves, policymakers and industry will have to work together to anticipate the power needs of the AI economy. Factors including future AI workloads, inhibitors to data center growth (land, energy availability, labor, supply chain, etc.), and the future policy landscape will meaningfully impact load growth associated with AI and its effect on American consumers. For example, as AI workloads shift from training to inference, so too will the energy needs of data centers powering future AI workloads. This will require that policies encourage innovation and flexibility amongst private stakeholders to adapt sufficiently and quickly. Today, we are working hand in hand with power providers and the IT industry to support the AI transition, and as the needs and use cases for AI evolve, Schneider will continue to robustly support these critical industries.

As new data centers are built and existing data centers are upgraded, both the private and public sector should look to this industry as a mechanism for stabilizing and expanding the grid. Today, data centers can be powered by converged energy systems that can assist not only in providing critical power to the data center, but providing stored power back to the grid, keeping costs low for consumers and preserving critical energy resources. These technologies are available and deployable today. Solutions internal to the data center like the use of liquid and innovative cooling and building management systems, and technologies and partnerships surrounding the data center like co-location of energy resources, extended battery energy storage systems (BESS), and uninterruptible power systems (UPSs) can help to stabilize the grid, shifting the conversation from “How will we power data centers?” to “How can data centers power and support their communities?”

The AI boom presents a clear moment for America to boldly modernize its energy ecosystem. AI workloads and the accelerated processor capacity they require are driving exponential growth in power consumption amongst AI data centers, requiring they be modern and upgraded to support the high density and power needs associated with evolving AI workloads. This window of opportunity – wherein AI data centers are being built and modernized – must be leveraged by policymakers and energy stakeholders to innovatively strengthen the U.S. grid and support American communities. Through a mix of energy resources, upgraded technologies capable of securely and reliably powering a variety of computing and storing needs, and increased flexibility to measure and handle complex and varying energy consumption needs, AI data centers can work with utility and infrastructure partners to meet their own energy needs while simultaneously assisting to bolster and power the grid.

Importantly, a one-size-fits-all approach to powering data centers and AI will fail in affording all data center operators the flexibility needed to address their role in rising demand and to assist in stabilizing the grid. Instead, a policy environment that fosters innovation and encourages adoption and deployment of new technologies across the grid will afford the data center market the agility it needs to grow the American AI economy. The Trump Administration can transform

the entire U.S. energy landscape by taking advantage of the historic trend in AI and data center investment – making energy more affordable and reliable for all consumers.

Increase Data Center Flexibility and Improve Load Management to Keep Energy Costs Low and Reduce Grid Strain

Responding to growing energy volatility will require decisive action, not only to secure and maintain ample energy supply, but also to leverage innovative technologies to measure, forecast, and optimize energy demand. Without demand-side interventions to manage load and supply-side innovations to boost production and mitigate losses from the delivery of electricity, consumers will continue to pay higher prices for energy. Therefore, we must overcome the barriers impeding our ability to leverage supply- and demand-side technologies and strategies that enable us to reduce losses and get the most out of the resources we have.

For data centers, the ability to leverage flexible, converged energy systems will allow them to optimize their energy use to make the most of critical and limited energy resources, helping to prevent rate hikes for consumers in their communities. While data centers have traditionally been viewed as substantial energy consumers, innovative technologies and strategies, including those incentivized through policy, are enabling an evolution in which data centers can be flexible loads that serve as grid stabilizers for enhanced reliability and dynamic load management. However, load variability of data centers can create uncertainty around the future of energy demand associated with AI.

Data Center Flexibility Strategies

In the coming years, the profile of AI data center demand will vary significantly based on whether data centers are enabling AI model training or inference. Currently, we estimate that AI workloads are split at 20 percent for training and 80 percent for inference, with baseload needs being relatively flat aside from seasonal variation.⁸ This trend may continue if new AI models are in constant development, making model training the primary driver of AI data center load. However, if AI usage overtakes training as the dominant load source, then demand curves would be more dependent on usage time and type.⁹ We estimate that AI workloads will shift to 15 percent training and 85 percent inference by 2028.¹⁰ While this variability creates uncertainty around future load profiles, it can also present an opportunity for load shifting where non-time-sensitive workloads can be scheduled to align with periods of lower grid demand to reduce grid strain, improve energy efficiency, and optimize energy costs.

⁸ Victor Avelar, Patrick Donovan, Paul Lin, Wendy Torell, and Maria A. Torres Arango, *The AI Disruption: Challenges and Guidance for Data Center Design*, Schneider Electric – Energy Management Research Center, Sep. 7, 2023, at 2.

⁹ Isabelle Riu, Dieter Smiley, Stephen Bessaparis, and Kushal Patel, *Load Growth is Here to Stay, but Are Data Centers?*, Energy and Environmental Economic, Inc., Jul. 2024 at 13; See Devin Hartman and Olivia Manzagol, *AI's Energy Footprint Warrants Markets, Not Panic*, R Street Institute, Sep. 26, 2024 at <https://www.rstreet.org/commentary/ais-energy-footprint-warrants-markets-not-panic/>.

¹⁰ Victor Avelar, Patrick Donovan, Paul Lin, Wendy Torell, and Maria A. Torres Arango, *The AI Disruption: Challenges and Guidance for Data Center Design*, Schneider Electric – Energy Management Research Center, Sep. 7, 2023, at 2.

Efficiency measures include devices (chips, servers, processors, power supplies), efficient arrangement of server racks, efficient cooling systems, and data center energy management.¹¹ True data center flexibility will incorporate this suite of approaches, and AI data centers may lend themselves to an enhanced range of options depending upon use case. Recent developments in computational load profiles, operational capabilities, and broader market conditions reveal greater potential for load flexibility from AI-driven workloads.¹² In contrast to the many real-time processing demands typical of conventional data center workloads like cloud services and enterprise applications, powering large language models and other machine learning algorithms is deferrable.¹³ This temporal flexibility allows for the strategic scheduling of training to ensure it's orchestrated to occur amongst more optimal conditions. Likewise, AI data centers may lend themselves to spatial flexibility, referring to the ability to dynamically distribute workloads across data centers in different geographic locations to optimize resource utilization and operational efficiency.¹⁴ In particular, AI training workloads are suited for this spatial flexibility given they are not as constrained by the same latency requirements as AI inference sites.¹⁵ This would also allow certain AI data centers to be located in less densely populated areas where land is more readily available and affordable, and the grid may be less congested. Therefore, the additional load from AI training can be more easily accommodated without overburdening the local grid, helping to prevent potential grid instability in high demand areas. This may also make it easier to co-locate data centers with generation resources that require more space such as solar or wind power.

A robust effort is already underway to explore the viability of these approaches and demonstrate the extent to which data center flexibility is possible. Schneider Electric is actively contributing to the proliferation of flexible data centers through participation in the Electric Power Research Institute's (EPRI) Data Center Flexible Load (DC Flex) Initiative. This collaborative effort, which involves a wide variety of stakeholders from utilities and power producers to chip providers and data center operators, aims to demonstrate how data centers can support and stabilize the electric grid while improving interconnection and efficiency. By integrating data centers as active participants in energy management, the initiative seeks to create a blueprint for stakeholders to adopt flexible load strategies like those already mentioned. However, exploring and demonstrating these capabilities is just the first step. As more of these concepts are proofed, the role of the government in facilitating lesson-sharing, best practice development, and policy guidance will be essential in ensuring replicability and scalability. The government can also play a catalytic role through policies that incent data centers to act as flexible resources including by helping to fill fundamental data and knowledge gaps.

Empower Data Center Flexibility through Technological Innovation

¹¹ Nora Wang Efram and Neal Elliott, *Turning Data Centers into Grid and Regional Assets: Considerations and Recommendations for the Federal Government, State Policymakers, and Utility Regulators*, ACEEE, Oct. 2024 at 6.

¹² Tyler Norris, Tim Profeta, Dalia Patino-Echeverri, and Adam Cowie-Haskell, *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in U.S. Power Systems*, Nicholas Institute for Energy, Environment, and Sustainability, Duke University, 2025 at 11/

¹³ *Id.*

¹⁴ *Id.*

¹⁵ See Devin Hartman and Olivia Manzagol, *AI's Energy Footprint Warrants Markets, Not Panic*, R Street Institute, Sep. 26, 2024 at <https://www.rstreet.org/commentary/ais-energy-footprint-warrants-markets-not-panic/>.

To fully realize the future potential of data centers as grid resources, data centers must be increasingly equipped with advanced solutions, such as grid-interactive technologies, that enable bidirectional capabilities, allowing energy to flow to and from the grid. An example of this are UPSs that not only draw power from the grid but also supply energy back during peak demand or grid disturbances. This capability helps mitigate growing load demand and reduce the energy burden on communities challenged by severe weather disasters.

Technologies like grid-interactive UPS systems represent just one piece of what may be included in a data center's converged energy system, allowing it to most effectively use available energy to meet computing demands. By allowing data centers to structure their own energy portfolio, policymakers can hasten commercialization of new and evolving energy resources and spur innovation. Other technologies that data centers can leverage to optimize their energy usage and serve as grid stabilizers include:

- Innovative cooling systems: The required density of AI clusters in data centers will require evolutions in cooling technologies. While conventional data center cooling systems may remain sufficient to manage less dense clusters and inference servers, a shift from air cooling to liquid – or other innovative – cooling systems will improve performance and reduce energy costs while safely managing the temperature within AI data centers.¹⁶ Innovation must continue to both ensure the responsible use of water for liquid cooling applications and source engineered fluids that can safely supplement water resources.¹⁷
- Microgrids: Microgrids can incorporate almost any form of generation, including natural gas, CHP, fuel cells, or solar, and many also integrate robust energy storage systems, enabling them to play a role across the entire range of various potential data center co-location configurations. The key piece of technology that defines a microgrid is its controller (the energy control center) which is comprised of hardware and software that allows the microgrid to separate from the main grid during power outages and switch to its on-site generators to supply continuous power to a data center¹⁸. In the face of outages or disruptions, the on-site system can take over automatically so that operations run seamlessly until grid power is restored. This functional capability reinforces the value of these assets as an additive solution in the co-located approach to data centers.
- Energy management systems (EMS): By using advanced controls, optimizing data center operations, and leveraging rapidly advancing AI for load forecasting in energy management systems, data center operators can also reduce data center energy consumption and integrate them with the local grid without sacrificing computational capacity.
- Distributed energy resource management systems (DERMS): DERMS are designed to manage diverse and dispersed distributed energy resources, both individually and in aggregate. They enable utilities / grid operators to efficiently manage consumer

¹⁶ Victor Avelar, Patrick Donovan, Paul Lin, Wendy Torell, and Maria A. Torres Arango, *The AI Disruption: Challenges and Guidance for Data Center Design*, Schneider Electric – Energy Management Research Center, Sep. 7, 2023, at 10.

¹⁷ *Id.* at 13.

¹⁸ Microgrid Knowledge and Data Center Frontier, *Data Center Microgrids: The Case for Microgrids at Data Centers*, 2022 at https://download.schneider-electric.com/files?p_Doc_Ref=22511651_en&p_enDocType=Brochure&p_File_Name=Microgrids+for+Data+Centers.pdf.

electricity demand and apply the capabilities of flexible demand-side energy resources. DERMS can be particularly effective in integrating BTM assets with the grid, providing dynamic load management and bidirectional power flows.¹⁹ For data centers to act as grid resources and stabilizers, DERMS can effectively manage both FTM and BTM systems, enabling dynamic load management and bidirectional power flows, thus enhancing grid stability and resilience.

- **Digital twins:** Digital twins are a virtual representation of a physical asset, process, or system that is continuously updated with real-time data. They provide real-time monitoring, predictive maintenance, energy optimization, and enhanced grid interaction, ultimately improving both data center operations and grid stability.²⁰

In the absence of regulatory guidance, most advancements in data center flexibility are being driven by voluntary private-sector initiatives. While these initiatives have spurred innovation and demonstrated the potential for data centers to act as flexible grid resources, they are inherently limited by the lack of standardized frameworks and incentives that only regulatory and policy guidance can provide. Regulatory guidance is essential to driving progress towards data center flexibility demonstration because it can establish clear, consistent standards for energy management, load balancing, and integration with the grid.²¹ Without such guidance, private-sector efforts may remain fragmented and insufficient to address the broader challenges of limited power supply, transmission and distribution capacity, and outdated infrastructure. Furthermore, regulatory policies can incentivize investments in advanced technologies and infrastructure upgrades, ensuring that data centers can reliably support national security and technological innovation. Therefore, private-sector initiatives alone cannot propel meaningful progress on this critical issue, balancing the competing priorities of American technological innovation and national security will require comprehensive policy frameworks that can guide and support the development of flexible, resilient data centers capable of meeting the demands of a rapidly evolving digital landscape.

Strengthen the Supply of Energy Resources and Products Supporting the AI Boom

From both the supply- and demand-side perspectives, the data center and AI boom represents a bold opportunity to leverage a key segment of America's economic engine to stabilize and power the grid. As a major consumer of electricity, data centers can – with the right policy and economic drivers – responsibly use the electricity they receive and generate to A) ensure sufficient availability of electricity for other utility consumers and B) provide excess and stored power back to the grid. This capability hinges on the availability and reliability of supply of existing and new energy resources as well as the availability of supply chains critical to building out data centers and power infrastructure.

¹⁹ ESG News, *Schneider Electric Accelerates Grid Modernization with Advanced DERMS, AI-powered Solutions, and Open Collaboration*, Feb. 29, 2024 at <https://esgnews.com/schneider-electric-accelerates-grid-modernization-with-advanced-derms-ai-powered-solutions-and-open-collaboration/>.

²⁰ AVEVA, *Digital twin software*, 2025 at <https://www.aveva.com/en/solutions/digital-transformation/digital-twin/>.

²¹ Milrose Consultants, *The Future of Data Center Development: Regulatory Trends Shaping the Industry*, Mar. 3, 2025 at <https://www.milrose.com/insights/the-future-of-data-center-development>.

Assess Availability and Corresponding Timelines to Power for “All-of-the-Above” Energy Resources

Today, several power resources for powering modern data centers – including nuclear, fuel cells, and natural gas turbines – are challenged by limited commercialization or development, high costs, permitting and regulatory delays, and/or supply chain backlogs. These energy generation sources, together with other distributed energy resources and renewables, can combine to provide data centers with a diverse, reliable energy portfolio capable of maintaining AI workloads. Increasingly, data center operators are looking to co-locate power generation sources on-site to reduce dependence on strained electrical grids, improve reliability, and support scalability. Combined with the varying maturity of available and newer energy resources, the lack of regulatory support for co-location challenges the ability of data center operators to access secure and reliable power generation.

Today, data centers leverage backup power systems – typically comprising diesel-generators and a UPS – to ensure continuity of service. As AI workloads increase and fluctuate, the inability of these systems to run continuously presents problems for AI data centers seeking consistent access to power. Leveraging and intelligently connecting various backup power technologies, including battery energy storage systems, with prime power sources is essential to data centers serving as grid stabilizers. Today, data centers are considering a myriad of different energy resources for prime and backup power, including:

Small Modular Reactors (SMRs): In the future, SMRs can serve as a stable, low-carbon energy source for high-performance computing facilities. Because SMRs are small and site-flexible, they can be deployed widely and built quickly on-site at an existing data center campus. Depending on energy consumption, the number of SMRs can be scaled to meet a data center’s evolving demands. Therefore, when deployed strategically and at scale, SMRs supply reliable, baseload power independent of the grid, increasing the resilience of energy systems while reducing energy costs for consumers. Challenges to the buildout of SMRs include regulatory hurdles, public concerns over nuclear fuel, and high upfront costs.²²

Natural Gas and Hybrid Solutions: High-efficiency natural gas plants, coupled with combined heat and power (CHP) systems, can offer reliable, lower-emission baseload power thereby making it an increasingly popular option for data center operators exploring co-location configurations. Schneider Electric believes that natural gas turbines will be the most commonly used prime power alternative in the coming years due to the technology’s maturity, cost, fuel flexibility, reliability, deployment speed, and scalability relative to other prime power options.²³ However, supply chain backlogs may challenge the short-term deployment of natural gas turbines as an immediate response to growing load demand from data centers.

On-Site Energy Storage: The incorporation of battery storage and distributed energy resources can help data centers manage peak demand and lower emissions. On-site energy storage systems

²² Patrick Donovan, *The Looming Power Crunch: Solutions for Data Center Expansion in an Energy-Constrained World*, Schneider Electric, Jan. 8, 2025 at 16.

²³ Patrick Donovan, *The Looming Power Crunch: Solutions for Data Center Expansion in an Energy-Constrained World*, Schneider Electric, Jan. 8, 2025 at 16.

like lithium-ion battery energy storage systems (Li-ion BESS) provide additional backup power, decrease reliance on diesel generators, and enable market participation and demand charge avoidance.²⁴ Li-Ion BESS, in particular, is a mature, safe technology capable of supporting runtimes of one to four hours.²⁵

Leveraging on-site energy storage in any form is necessary to empower data centers to truly stabilize the grid and power their own operations. Without sufficient on-site energy storage capacity, a data center will not be able to remove their load from the grid as needed or directed. Furthermore, an Energy-as-a-Service (EaaS) model enables enterprises to integrate on-site distributed energy without upfront capital investment, helping to accelerate deployment.

Fuel Cells: In regions with limited grid power, fuel cells may serve as an alternative for prime power to data centers. Challenges with deploying fuel cells broadly include cost, longer startup times, the need for additional infrastructure, and difficulties in following larger swings in load demand. Additionally, the constrained supply of green hydrogen challenges fuel cell adoption.²⁶

Flexible policy that enables the interconnection of these resources both in front-of- and behind-the-meter, as well as spurs private sector innovation to speed development of new American energy resources, will benefit all American energy consumers.

Bolster Supply Chains Supporting the AI Economy

Complex supply chains – from raw materials to advanced technological components – are necessary to support the buildout and success of America’s AI economy. Not only should the Administration consider the reliability of supply chains supporting infrastructure internal to data centers, but it must also consider implications of supply chain challenges that impact the U.S. grid. Shortages in supply of semiconductor components, servers, cooling systems and components, networking equipment, back-up systems, and storage devices are likely to impact the development and expansion of data centers in the United States. Outside of the data center, supply chains for critical minerals (including copper, lithium, metals etc.) and critical grid components (including transformers, switchgear, electrical steel, and other power distribution equipment) face issues with availability, resiliency, and backlogs.

It is crucial the Trump Administration continue working to mitigate these supply chain gaps by building domestic capacity and working with trading partners to fortify trusted supply chains that are critical to the American AI economy.

The concept of modern data centers serving as grid stabilizers presents an opportunity for the Administration to leverage the AI boom to power America’s energy independence. By making the most of our new and existing energy resources – including via growing markets and

²⁴ Steven Carlini, *Data centers and the power squeeze – is this an opportunity for innovation?*, Data Center Dynamics, Aug. 27, 2024 at <https://www.datacenterdynamics.com/en/opinions/data-centers-and-the-power-squeeze-is-this-an-opportunity-for-innovation/>; See Patrick Donovan, *Understanding BESS: Battery Energy Storage Systems for Data Centers*, Schneider Electric, Nov. 13, 2024.

²⁵ Patrick Donovan, *The Looming Power Crunch: Solutions for Data Center Expansion in an Energy-Constrained World*, Schneider Electric, Jan. 8, 2025 at 13.

²⁶ *Id.* at 16

segments (inclusive of data centers, re-shored manufacturing, etc.), and investing in the strength of relevant supply chains, the Administration can ensure America leads the world in energy affordability, innovation, and reliability.

Drive Investment in the Future of AI in America

American innovation and tenacity have propelled the AI boom, and sustaining that trajectory requires close partnership between the public and private sectors to work towards investments and innovations that are most meaningful and impactful to the U.S. economy.

Certainty and consistency in tax policy, strong investments in research and development (R&D), and a rational regulatory landscape that considers differences in technology by application and sector will continue to spur private sector investment at the pace necessary to guarantee American AI leadership.

Support Research and Development

America's leadership in innovation through R&D is particularly crucial to our national security and economic competitiveness. Today, Schneider Electric plays a role through our commitment to advancing energy-efficient AI infrastructure through strategic R&D initiatives. In 2024, we invested \$2.4 billion globally to support R&D efforts, including the establishment of new data center and microgrid testing labs at our [Innovation Hub](#) in Andover, Massachusetts, just outside our U.S. headquarters in Boston.²⁷ With this added investment, we have over 40 labs in Andover, including a Power Distribution Unit (PDU) lab to test high-powered voltage systems that are designed and developed for the AI data center market. Innovations in liquid cooling, AI-optimized power management, and next-generation semiconductor efficiency are essential to reducing the energy demands of AI, and this is an effort that we've committed to through our investment in liquid cooling technologies.

Schneider Electric also leads AI R&D through our dedicated global [AI Hub](#) comprised of over 350 experts. The AI Hub serves as a global center of excellence utilizing AI technologies to transform and optimize energy management and industrial automation. Some of the examples include energy flexibility optimization for Microgrids and homes, HVAC optimization for buildings, and predictive maintenance for Electrical Distribution equipment. When the AI Hub was launched in 2021, GenAI technology was not yet prevalent the way it is today. However, with the emergence of these new tools, we created a dedicated GenAI core team to work closely with internal stakeholders and external partners, like Microsoft and Amazon, to detect early application opportunities that can drive efficiency gains and help achieve sustainable growth. For instance, our [Resource Advisor Copilot](#) leverages Large Language Model (LLM) technology through Microsoft Azure OpenAI to equip customers with enhanced data analysis, visualization, decision support, performance optimization, and the ability to seamlessly process intricate

²⁷ Schneider Electric, *Schneider Electric Opens New Data Center and Microgrid Testing Labs at Global R&D Center in Massachusetts*, Feb. 27, 2025 at <https://www.se.com/us/en/about-us/newsroom/news/press-releases/schneider-electric-opens-new-data-center-and-microgrid-testing-labs-at-global-r-d-center-in-massachusetts-67b4ab53ee870308eb057e31>.

industry knowledge and Resource Advisor system information.²⁸ This demonstrates our commitment to being on the cutting-edge of innovation and delivering the benefits of these technological advancements to our customers, including the development and application of AI solutions.

Partnerships across the private sector and public sector will be critical to the future of AI in America. To that end, continued investments in the Department of Energy's national laboratories and support for private sector R&D – including through immediate expensing – are critical to making America the world's premier AI R&D engine.

Determinations of Risk Should be Use- and Sector-Specific

Allowing developers and deployers of AI to self-assess risk for their unique use cases and organize their risk management processes creates flexibility for organizations to appropriately manage risk for their unique use cases. While organizations may have diverse ways to carry out a risk assessment, leveraging voluntary private sector-led consensus standards and voluntary tools like the National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF) can provide a uniform language for AI risk management throughout the lifecycle.

For critical infrastructure use cases, it is critical the government continue working with the private sector to properly assign risk and regulate AI use. Efforts like the Department of Energy's initial [risk assessment](#) on the use of AI in critical energy infrastructure are important steps toward building a clearer common understanding of the risks and rewards of AI's use in critical infrastructure. Developed in partnership with Lawrence Livermore National Laboratory (LLNL), energy sector owners and operators, and other technical subject matter experts, this effort is also an important example of the public-private cooperation needed to advance safe and secure AI in critical infrastructure. As a long-standing partner and the first participating manufacturer to test products used in the U.S. grid through DOE's Cyber Testing for Resilient Industrial Control Systems ([CyTRICS](#)) program, Schneider Electric recognizes the importance of industry collaboration and deeply values opportunities to partner with the U.S. government to secure American critical infrastructure.

Advance Responsible AI Cybersecurity Policies

Schneider Electric views AI as both an opportunity and a challenge in maintaining its cybersecurity posture. This dual perspective underscores our proactive approach to integrating AI into our operations and products. For example, by embedding AI into threat detection systems, Schneider Electric enhances its ability to identify and mitigate cyber risks in real time—a critical capability in today's rapidly evolving digital landscape. At the same time, Schneider Electric recognizes that the security of AI is paramount when that AI is used to control and

²⁸ Schneider Electric, *Building Sustainability's Digital Future with EcoStruxure Resource Advisor Copilot: Schneider Electric's Latest AI Advancement*, Aug. 29, 2023 at <https://www.se.com/ww/en/about-us/newsroom/news/press-releases/building-sustainability%E2%80%99s-digital-future-with-ecostruxure%E2%84%A2-resource-advisor-copilot-schneider-electric%E2%80%99s-latest-ai-advancement-64eca1cf6053e3d1d90ab96e>.

operate critical infrastructure. Schneider Electric's Secure Development Lifecycle incorporates myriad controls to make sure AI is securely and responsibly developed for integration into our energy efficiency and industrial automation products. Led by our global AI Hub, our AI efforts are implemented with responsible AI practices, ensuring deployment is safe, trustworthy, and ethical.

At the same time, Schneider Electric recognizes that the security of AI is paramount when that AI is used to control and operate critical infrastructure. Schneider Electric's Secure Development Lifecycle incorporates myriad controls to make sure AI is safely, securely, and responsibly developed for integration into our energy efficiency and industrial automation products.

Conclusion

As a key infrastructure provider from chip to data center to grid, Schneider Electric hopes the Trump Administration will leverage our work and expertise as a resource as it develops its AI Action Plan and future policies around AI and energy. Our recent investments in the U.S. are paving the path for a stronger, more resilient domestic supply chain supporting data center and grid buildout and we look forward to continuing to partner with the Trump Administration to make the most of these efforts.

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